

Jagraj Gill

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Education

Simon Fraser University, Bachelor of Science in Computing Science

Dec 2027

Skills

Languages: C, C++, Python, Java, JavaScript, HTML, CSS, x86 Assembly

Frameworks: React, Node.js, Express.js, REST APIs, GraphQL, FastAPI, CI/CD, Agile

Databases & Tools: Git, MongoDB, MySQL, SQLite, AWS S3, Linux, ESP-IDF

Embedded Systems: Microcontrollers, ARM-Cortex, RTOS, I2C, UART, TCP/IP

Experience

Software Engineer Intern, Savi Finance – Toronto, ON

May 2025 – Dec 2025

- Developed an AI-powered task scheduling system that ranked tasks by priority, **reducing planning time by 10%**
- Built shared modules across React and React Native with a MongoDB backend for a fintech SaaS platform serving **1000+ users, shortening release cycles by 15%**
- Integrated an image upload pipeline using AWS S3 and GraphQL, adding custom preview support and reducing image load failures by **10%**
- Refactored login UI and validation using React and custom hooks, **cutting user login errors by 45%**

Hackathon, Storm Hacks 2024 – Burnaby, BC

May 2024

- Designed REST APIs with Node.js and Express.js to handle real-time data synchronization, **improving system response time by 30%**
- Tested and debugged backend endpoints, helping deliver a working prototype within 24 hours

Projects

AI Study Pal - Python, FastAPI, Gemini API, HTML, CSS, JavaScript

[GitHub](#)

- Built an LLM-powered pipeline that converts raw PDF documents into structured flashcard datasets, automating study material generation from unstructured text
- Designed a strict JSON schema for AI-generated flashcards, improving output consistency and enabling reliable parsing
- Implemented validation logic in Python to verify data structure, field integrity, and content quality, preventing malformed or incomplete AI responses from entering the system
- Developed an interactive flashcard interface with flip animations and navigation controls, enabling real-time study directly from AI-generated content

IoT Climate Monitoring Dashboard - C, ESP-IDF, FastAPI, SQLite, JavaScript, HTML, CSS

[GitHub](#)

- Built a full-stack IoT monitoring system integrating ESP32 firmware, FastAPI REST endpoints, SQLite storage, and a JavaScript dashboard for real-time sensor visualization
- Developed a FastAPI backend that ingested structured sensor data from ESP32 firmware over Wi-Fi, storing readings in SQLite and serving live updates to a JavaScript dashboard
- Built a 24-hour data pipeline with SQLite and REST endpoints, enabling live readings, device status, and trend visualization

Investment Tracker - Java, Swing, JFreeChart

[GitHub](#)

- Built a Java desktop application with a Swing GUI to render portfolio growth across multiple assets using JFreeChart graphs
- Improved reliability through exception handling and input validation, ensuring consistent behavior for dynamic entries
- Implemented portfolio calculations using prices, quantities, and purchase history to compare performance across holdings

Wireless 4-DOF Robotic Arm - ESP32, C, Embedded Systems

[GitHub](#)

- Built the full software stack for a 4-DOF robotic arm on ESP32, spanning embedded C firmware, analog signal processing, and wireless communication between controllers
- Developed embedded C firmware that processed analog sensor inputs and drove PWM signals across 4 servo joints for real-time actuator control
- Implemented wireless communication between 2 ESP32 devices, enabling synchronized multi-joint motion control through real-time data transmission
- Structured modular control logic to separate input processing, wireless communication, and actuator control